

Project Summary

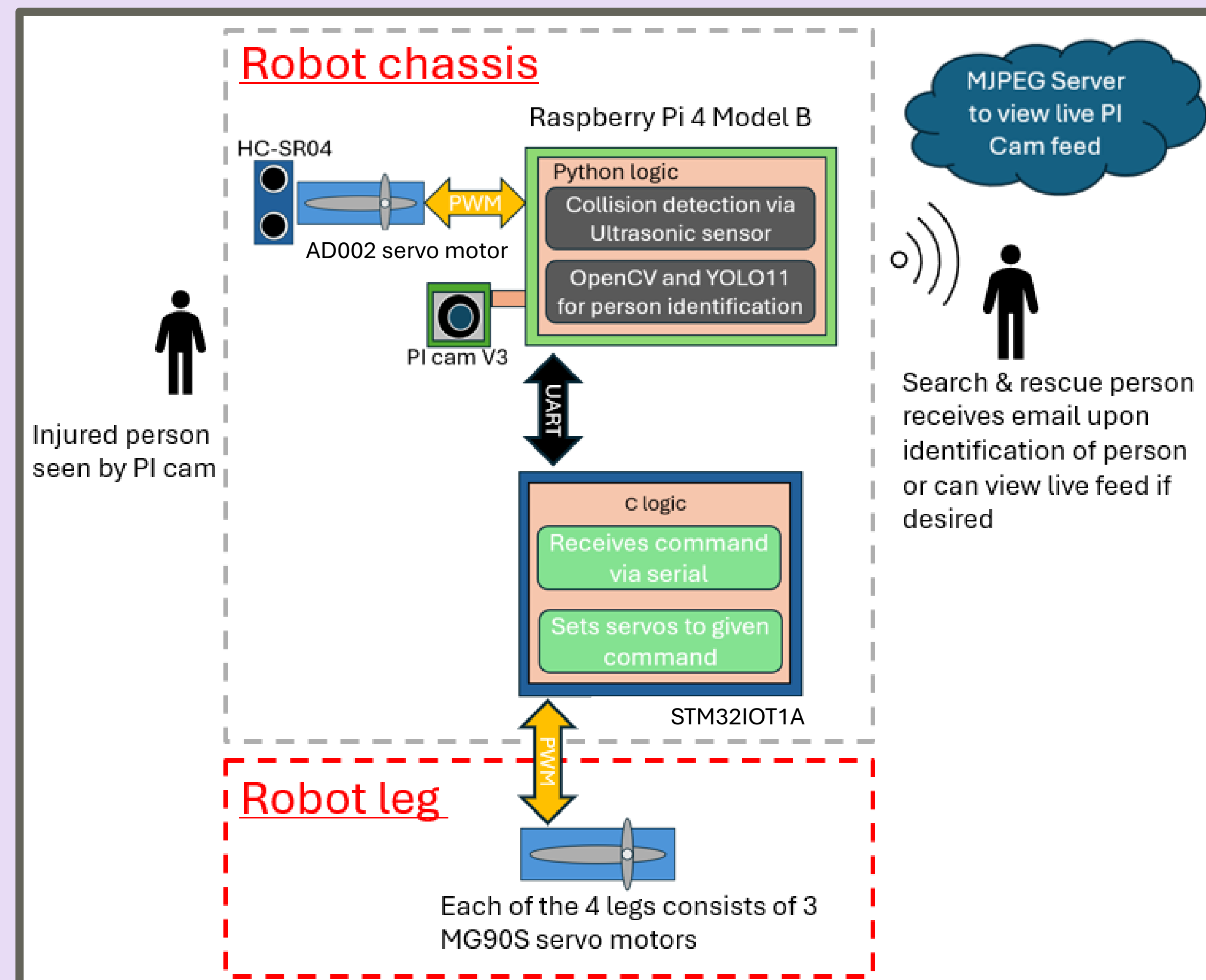
The Search & Rescue Quadraped (SaRQ) is designed to be deployed in search & rescue scenarios, detecting injured persons and alerting search & rescue operators.

The SaRQ consists of a Raspberry Pi (RPI) communicating with an STM32IOT1A (STM32) to control MG90S servo motors to create movement.

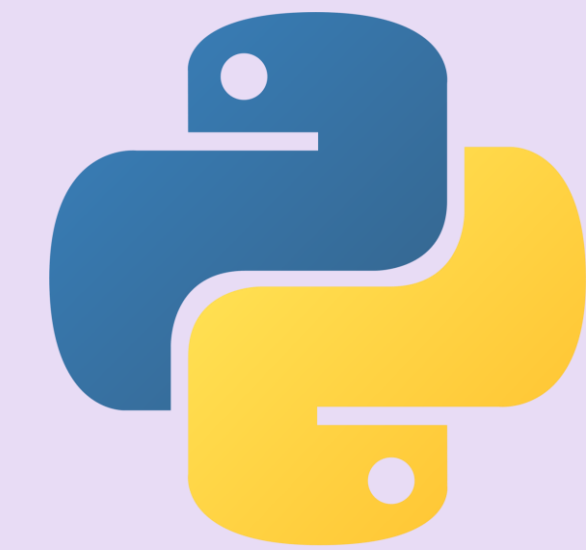
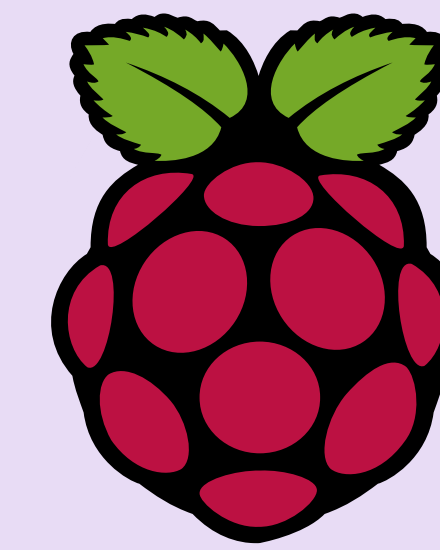
The RPI is connected to a HC-SR04 ultrasonic sensor for collision detection and a Pi camera V3 running the you only look once (YOLO) detection algorithm. If YOLO detects a person, an email is sent to a search and rescue person.

The SaRQ's live feed through a motion jpeg (MJPEG) server is also available.

Architecture Diagram



Technologies Used



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Hardware Design

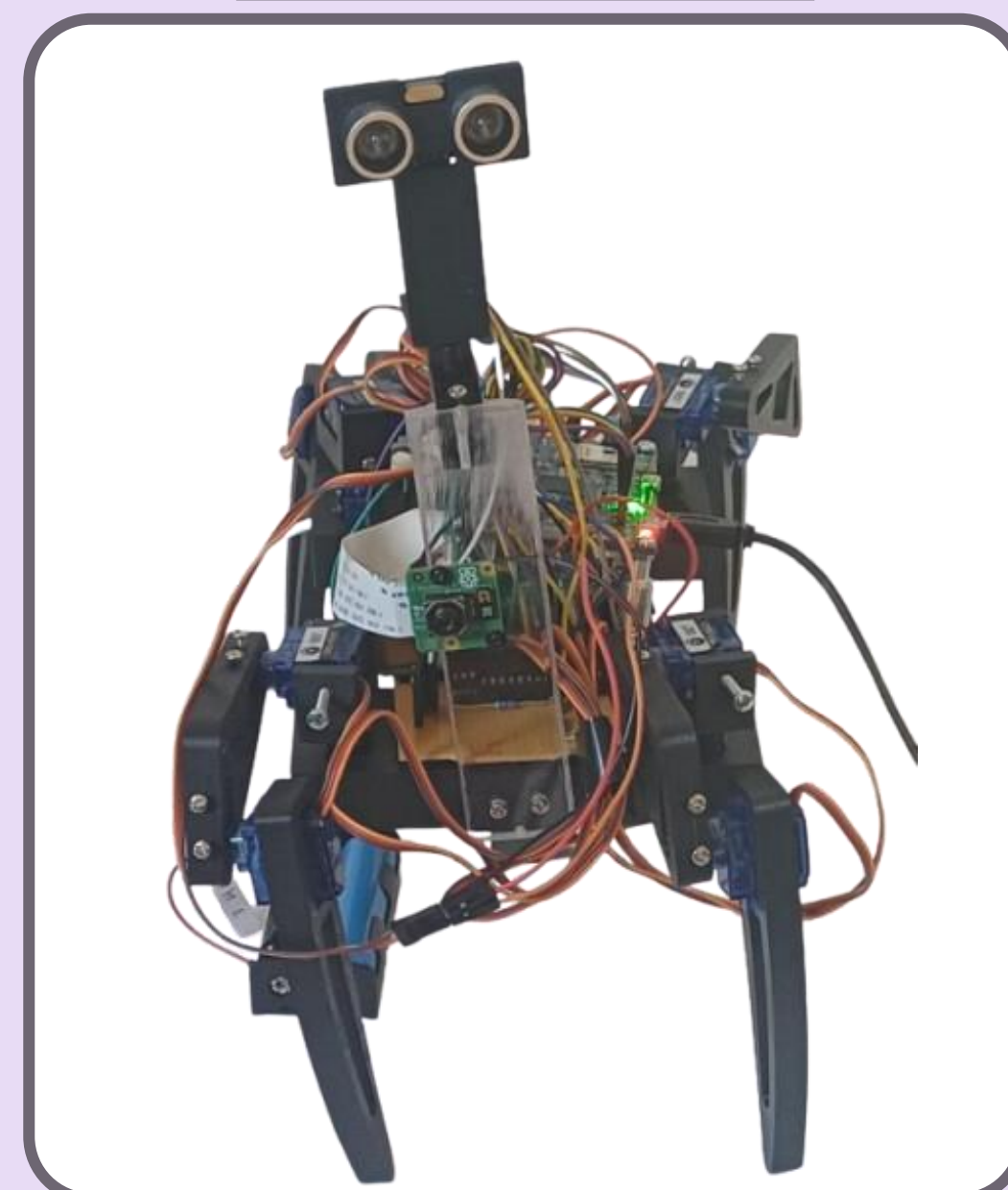
The design of the legs, chassis and holder for the HC-SR04 were drawn and designed in Onshape.

A Perspex mount holds the Pi camera on the front, along with a servo motor that rotates the HC-SR04.

A series of M3 screws hold the SaRQ, where the holes were drilled and tapped, providing a better grip.

A brown stripboard sits beneath the RPI and STM32, containing a series of headers to help manage the number of servo motor connections, aiding wire management.

The SaRQ



Results

If the HC-SR04 doesn't detect an obstacle in front, the RPI tells the STM32 over serial to step forward. The STM32 controls the legs via servo motors and moves when told.

If an obstacle is detected, the HC-SR04 rotates via an SG90 servo motor, taking readings on the left and right of the SaRQ for the RPI to decide the best direction to move.

A live feed is shown over an MJPEG server hosted on the RPI. YOLO can detect a user, and an email is sent automatically of the person detected to a search & rescue person.